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Basic Python Programming

Q: 1 Write a Python program which accepts a sequence of comma-separated numbers from user and generate a list and a tuple with those numbers. Sample data : 3, 5, 7, 23 Output : List : ['3', ' 5', ' 7', ' 23'] Tuple : ('3', ' 5', ' 7', ' 23')

```
# Accept input from the user
values = input("Input some comma separated numbers: ")

# Split the string by the comma delimiter to create a list
list_data = values.split(",")

# Convert the list into a tuple
tuple_data = tuple(list_data)

# Print the results
print('List : ', list_data)
print('Tuple : ', tuple_data)
```

Q: 2 Write a Python program to print the documents (syntax, description etc.) of Python built-in function(s).

Sample function : abs() Expected Result : mat abs(number) -> number Return the absolute value of the argument.

```
# Example using a specific function directly
print("--- Direct Access ---")
print(abs.__doc__)

# Example allowing user input
print("\n--- Interactive Access ---")
func_name = input("Enter a built-in function name (e.g., len, max, input): ")

try:
    # eval() converts the string input (e.g., "len") into the actual function
object
    func_object = eval(func_name)

    # Access and print the __doc__ attribute
    print(func_object.__doc__)
except NameError:
    print(f"Error: '{func_name}' is not a recognized function.")
```

Q:3. Write a Python program to calculate number of days between two dates. Sample dates : (2014, 7, 2), (2014, 7, 11) Expected output : 9 days

```
from datetime import date

# Define the two dates: date(year, month, day)
first_date = date(2014, 7, 2)
last_date = date(2014, 7, 11)

# Subtracting dates returns a 'timedelta' object
delta = last_date - first_date

# Access the 'days' attribute of the delta object
print(f"{delta.days} days")
```

Q:4. Write a Python program to create a histogram from a given list of integers.

```
def histogram(items):
    for n in items:
        # Multiply the character by the number to create the bar
        print('*' * n)

# Test Data
histogram([2, 3, 6, 5])
```

Q:5. Write a Python program to print out a set containing all the colors from color_list_1 which are not present in color_list_2. Test Data : color_list_1 = set(["White", "Black", "Red"]) color_list_2 = set(["Red", "Green"]) Expected Output : {'Black', 'White'}

```
# Define the two sets
color_list_1 = set(["White", "Black", "Red"])
color_list_2 = set(["Red", "Green"])

# Calculate the difference
unique_colors = color_list_1.difference(color_list_2)

print(unique_colors)
```

Q:6. Write a Python program to count the number occurrence of a specific character in a string.

```
s = "The quick brown fox jumps over the lazy dog."
char_to_count = "o"
```

```
count = 0

for char in s:
    if char == char_to_count:
        count = count + 1

print(f"The character '{char_to_count}' appears {count} times.")
```

Q:7. Write a Python program to sort three integers without using conditional statements and loops.

```
# Accept inputs
x = int(input("Input first number: "))
y = int(input("Input second number: "))
z = int(input("Input third number: "))

# Find the smallest and largest numbers
a1 = min(x, y, z)
a3 = max(x, y, z)

# (Sum of all) - (Smallest) - (Largest) = Middle
a2 = (x + y + z) - a1 - a3

print("Sorted numbers:", a1, a2, a3)
```

Q:8. Write a Python program to get the size of an object in bytes.

```
import sys

# Define a few different types of objects
str_val = "Python"
int_val = 100
list_val = [1, 2, 3]

# sys.getsizeof() returns the size in bytes
print(f"Memory size of '{str_val}' (String): {sys.getsizeof(str_val)} bytes")
print(f"Memory size of '{int_val}' (Integer): {sys.getsizeof(int_val)} bytes")
print(f"Memory size of '{list_val}' (List):   {sys.getsizeof(list_val)}
bytes")
```

Q:9. Write a Python program to access and print a URL's content to the console.

```
# The 'requests' way to fetch a web page.
import requests

response = requests.get("https://www.python.org/")
print(response.text[:500])
```

Q:10. Write a Python program to get numbers divisible by fifteen from a list using an anonymous function.

```
num_list = [45, 55, 60, 37, 100, 105, 220]

# use filter() with a lambda function
result = list(filter(lambda x: (x % 15 == 0), num_list))

print("Numbers divisible by 15:", result)
```

Q:11. Write a Python program to extract single key-value pair of a dictionary in variables.

```
# Sample dictionary
d = {"Item": "Laptop", "Price": 1000}

key, value = next(iter(d.items()))

print(f"Key:   {key}")
print(f"Value: {value}")
```

Q:12. Write a Python function to find the maximum and minimum numbers from a sequence of numbers. Note: Do not use built-in functions.

```
def find_max_min(data):
    #  Handle edge case: Empty list
    if len(data) == 0:
        return None, None

    # We assume the first number is both the largest and smallest

    max_val = data[0]
    min_val = data[0]

    for num in data:
        # Check for new maximum
        if num > max_val:
            max_val = num

        # Check for new minimum
        elif num < min_val:
            min_val = num

    return max_val, min_val

numbers = [0, 10, 15, 40, -5, 42, 17, 28, 75]
largest, smallest = find_max_min(numbers)
```

```
print("Maximum Number:", largest)
print("Minimum Number:", smallest)
```